

XINGTAI, China

WMO: 537980

Lat: 37.1808N

Lon: 114.3589E

Elev: 184

StdP: 99.13

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-5.9	-23.9	0.4	0.3	-21.4	0.6	0.9	7.2	1.9	6.0	1.6	1.5	20	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	35.9	22.3	34.3	22.7	33.0	23.0	27.3	31.6	26.5	30.6	25.7	29.7	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.1	30.0	25.4	21.0	29.5	24.5	20.0	28.9	88.3	32.0	84.6	30.7	81.0	29.7	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.8	4.8	-9.8	38.7	1.6	1.5	-10.9	39.8	-11.8	40.7	-12.7	41.5	-13.8	42.6		
			-11.1	28.6	1.5	0.6	-12.2	29.0	-13.1	29.4	-13.9	29.8	-15.0	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	-0.4	3.2	9.8	16.3	22.1	26.6	27.8	26.3	22.0	16.0	7.6	1.4
	DBStd	10.47	3.11	3.80	4.38	3.97	3.38	3.03	2.63	2.54	3.05	3.51	4.27	3.11
	HDD10.0	938	323	193	56	3	0	0	0	0	0	2	96	265
	HDD18.3	2305	581	424	266	83	8	0	0	6	89	323	523	
	CDD10.0	2747	0	2	51	193	376	498	552	507	361	187	23	0
	CDD18.3	1073	0	0	2	23	126	248	293	249	117	15	0	0
	CDH23.3	10923	0	1	21	193	1116	2800	3414	2466	808	102	3	0
	CDH26.7	4521	0	0	4	44	394	1334	1573	938	218	16	0	0
Wind	WSAvg	2.0	1.8	2.0	2.4	2.6	2.5	2.3	1.9	1.8	1.8	1.7	1.7	1.7
Precipitation	PrecAvg	521	4	8	12	25	41	57	150	122	56	25	18	4
	PrecMax	806	18	43	55	53	108	113	371	351	108	102	65	23
	PrecMin	349	0	0	0	5	4	13	49	34	4	3	0	0
	PrecStd	121	5	9	16	16	25	24	80	72	30	23	20	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	16.8	26.0	30.9	35.6	38.4	37.9	35.3	32.4	28.6	22.3	13.1
		MCWB	3.9	7.4	12.5	16.5	18.3	21.0	23.6	25.8	22.2	16.4	11.2	4.5
	2%	DB	8.6	13.8	22.4	27.7	32.2	36.1	35.8	33.5	30.6	26.0	18.8	10.3
		MCWB	1.7	5.8	11.7	16.3	18.5	20.6	23.8	25.2	20.7	15.1	10.2	3.3
	5%	DB	6.5	11.6	19.7	25.5	30.5	34.5	34.3	32.3	29.2	24.0	16.4	8.2
		MCWB	0.7	4.4	10.2	15.1	18.0	20.5	24.5	24.6	20.0	14.5	9.1	2.3
	10%	DB	4.7	9.5	17.0	23.4	28.8	32.9	32.9	31.1	27.7	22.1	14.1	6.3
		MCWB	-0.4	3.4	8.8	14.3	17.7	20.7	24.5	24.3	19.7	14.2	8.0	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	9.0	15.5	20.6	22.7	25.3	28.5	28.1	25.5	19.3	13.0	5.6
		MCDB	11.1	14.3	21.7	26.5	28.7	32.1	33.2	32.6	29.7	24.4	19.1	10.9
	2%	WB	2.3	6.7	12.7	18.6	21.5	24.1	27.6	27.3	23.7	17.8	11.4	4.0
		MCDB	7.5	11.8	20.4	24.5	28.0	30.8	32.0	31.5	27.7	22.0	16.4	8.3
	5%	WB	1.0	5.2	11.0	17.0	20.5	23.2	26.8	26.5	22.2	16.7	10.2	3.0
		MCDB	5.7	10.2	18.0	23.0	27.2	29.8	31.0	30.4	26.5	21.0	15.1	7.2
	10%	WB	0.0	3.9	9.5	15.6	19.5	22.4	26.1	25.7	21.1	15.7	9.0	1.9
		MCDB	4.1	9.0	16.0	21.3	26.1	29.1	30.2	29.4	25.6	19.8	13.3	5.6
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.9	9.1	9.5	9.6	9.6	7.7	7.3	8.5	9.0	8.2	7.1
		MCDBR	10.5	11.6	12.6	12.9	12.7	12.3	10.1	8.9	10.6	12.2	12.1	10.5
		MCWBR	6.3	6.6	6.4	6.0	5.5	3.7	3.3	3.4	3.9	4.8	6.0	6.1
	5% WB	MCDBR	9.6	10.3	11.6	11.1	10.3	9.6	7.7	7.5	8.2	8.8	10.6	9.1
		MCWBR	6.1	6.3	6.5	6.3	5.4	3.6	3.5	3.6	3.9	4.4	5.7	5.6
Clear-Sky Solar Irradiance	taub	0.498	0.577	0.579	0.619	0.651	0.796	0.782	0.759	0.701	0.647	0.529	0.480	
	taud	1.811	1.671	1.694	1.651	1.635	1.439	1.506	1.528	1.583	1.635	1.795	1.868	
	Ebn at Noon	630	629	688	693	680	586	590	588	588	564	593	614	
	Edh at Noon	161	210	224	246	254	310	288	276	248	215	163	143	
All-Sky Solar Radiation	RadAvg	2.42	2.98	4.33	5.18	5.73	5.25	4.62	4.48	3.89	3.19	2.54	2.25	
	RadStd	0.26	0.46	0.41	0.39	0.46	0.44	0.29	0.44	0.38	0.37	0.42	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page